REAL-TIME TRAFFIC ANALYTICS USING AZURE

Project Planning & Management | DEPI R5 Graduation Project

Project Proposal

Real-Time Traffic Analytics Using Azure

Overview

The project demonstrates a near-real-time data-engineering pipeline using recent data from Finland's Digitraffic APIs. Python producers poll selected APIs and publish events to Kafka. Python consumers persist source records to PostgreSQL. dbt Core transforms raw data into Staging and Gold models for Power BI.

Problem and Objectives

Traffic information is distributed across endpoints and formats. The solution provides a repeatable way to ingest, validate, transform, and present the latest available information.

- Ingest recent data from validated Digitraffic endpoints.
- Demonstrate Kafka-based event streaming.
- Preserve source payloads and ingestion metadata in PostgreSQL.
- Build tested dbt models and a Power BI Desktop dashboard.
- Containerize services with Docker and assess deployment on an Azure VM.

Scope

In scope: API ingestion, Python producers/consumers, Kafka, PostgreSQL Raw layer, dbt Staging/Gold, Power BI Desktop, Docker, and Azure VM deployment assessment.

Out of scope: traffic control, guaranteed congestion prediction, direct road-control integration, and historical trend analysis using a pre-existing historical dataset. Raw snapshots may be retained for traceability/reprocessing under a retention policy.

Proposed API Scope

All paths and schemas must be validated against the current official OpenAPI before implementation.

- TMS: /api/tms/v1/stations/data, /api/tms/v1/stations, /api/tms/v1/sensors
- Maintenance: /api/maintenance/v1/tracking/routes/latest, /api/maintenance/v1/tracking/tasks, /api/maintenance/v1/tracking/routes
- Traffic messages V1: /api/traffic-message/v1/locations, /api/traffic-message/v1/locations/types, /api/traffic-message/v1/area-geometries
- Traffic messages V2: /api/traffic-message/v2/roadworks, /api/traffic-message/v2/traffic-announcements, /api/traffic-message/v2/traffic-data/datex2-3.7.xml
- Variable signs: /api/variable-sign/v1/signs
- Weather: /api/weather/v1/stations/data, /api/weather/v1/forecast-sections/forecasts, /api/weather/v1/stations, /api/weather/v1/forecast-sections

Official references: https://www.digitraffic.fi/en/support/instructions/ ; https://tie.digitraffic.fi/swagger/openapi.json ; base URL: https://tie.digitraffic.fi.

Architecture

Digitraffic APIs → Python Producers → Kafka → Python Consumers → PostgreSQL Raw → dbt Core (Staging → Gold) → Power BI Desktop. Docker packages services. Kafka and PostgreSQL are planned for an Azure VM; Power BI Desktop initially runs locally. Power BI Service is optional and depends on licensing, connectivity, and refresh constraints. Kafka does not poll APIs; Python producers do.

Data Layers

- Raw: preserve original JSON/XML payloads, source endpoint, ingestion timestamp, source update timestamp if supplied, and suitable identifiers/hashes.
- Staging: standardize types/names/timestamps and apply endpoint-specific validation and version rules.
- Gold: current, analytics-ready entities and metrics.

Differentiate identical re-ingestion, new observations with unchanged values, and genuine source updates. Do not use global deduplication rules.

Feasibility and Success

Feasibility depends on API cadence/availability, payload volume, VM resources, network access, and Power BI connectivity. Evaluate VM size and cost; do not assume a small VM supports the entire stack. Freshness of 5–10 minutes is a conditional target only when source cadence and platform limits permit. KPI targets remain unverified until measured.

Deliverables and Assumptions

Deliverables: source inventory, architecture, ingestion/streaming code, PostgreSQL schema, dbt models/tests, dashboard, Docker/deployment instructions, test evidence, KPI results, and final documentation. Confirm endpoint terms, paths, schema, cadence, secrets management, and scope before implementation.

Project Plan

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Planning Basis

This is a preliminary six-week plan. Align it with the actual DEPI deadline and available hours. Build one complete source-to-dashboard MVP before expanding endpoint coverage.

Work Breakdown

1. Requirements and source validation: confirm dashboard questions, endpoint paths, payloads, cadence, limits, and identifiers.

2. Architecture/data design: define Kafka topics, message contracts, PostgreSQL raw storage, retention, and dbt models.

3. Ingestion/streaming: build resilient Python API clients/producers, Kafka publishing, consumers, retries, and logging.

4. Storage/transformation: persist payloads and metadata; implement dbt Staging/Gold and data-quality tests.

5. Visualization: define measures, build Power BI Desktop report, reconcile calculations.

6. Integration/deployment: Dockerize services, test locally, assess Azure VM cost/resources, and document run/recovery steps.

Preliminary Timeline

Week	Activities	Milestone
Week 1	Requirements gathering, API review, and MVP endpoint selection.	**M1:** Scope defined and endpoint shortlist finalized.
Week 2	PostgreSQL and Kafka design, plus development of the producer skeleton.	**M2:** Design reviewed and first endpoint successfully publishing messages.
Week 3	Develop API producers and Kafka consumers, and implement raw data persistence.	**M3:** End-to-end source-to-raw data flow demonstrated.
Week 4	Develop dbt models, implement tests, and perform data reconciliation.	**M4:** Gold models implemented and tested.
Week 5	Build the Power BI dashboard and validate calculations against database results.	**M5:** Dashboard completed and measures reconciled.
Week 6	Integrate services using Docker, assess deployment requirements, test recovery procedures, and finalize documentation.	**M6:** Reproducible end-to-end demo completed and project handover ready.

Dependencies and Tracking

API validation precedes producer finalization. Raw keys precede dbt models; Gold definitions precede dashboard measures. Power BI Service is optional and must not block the Desktop MVP. Track owner, due date, status, acceptance criteria, and evidence in GitHub issues/board. Complete tasks only when committed, documented, and tested. If time is constrained, prioritize a tested MVP over broad unverified endpoint coverage.

Task Assignment & Roles

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Member	Primary Role	Responsibilities
Amr Mahmoud	Team Lead, Architecture & Integration	Coordinate scope and milestones, maintain architecture, review contracts, integrate the pipeline, manage risks, and prepare final documentation.
Mohamed Ibrahim	API Ingestion & Python Producers	Validate APIs, implement clients, parsing, retries, logging, and Kafka producers; document API update cadence and payloads.
Mohsen Khalaf	Testing & Data Validation	Perform smoke tests, verify sample records and row counts, document defect evidence, and retest fixes while keeping the workload moderate.
Mohamed Saleh	PostgreSQL & dbt	Design the raw schema and keys, manage payload metadata, develop Staging and Gold models, implement dbt tests, and perform reconciliation.
Basant Magdy	Power BI & Reporting	Define dashboard questions and measures, build Power BI Desktop pages, reconcile calculations, and document refresh limitations.
Kerolos Adly	Docker & Azure Deployment	Configure Docker and service networking/environment variables, assess VM sizing and costs, and document deployment and recovery procedures.

Shared Responsibilities

All members use Git branches/review, keep secrets out of source control, document assumptions/schema changes, update task status, and raise blockers early.

Handoffs

Mohamed Ibrahim provides endpoint contracts and producer messages; Amr and Mohamed Saleh agree on message/storage contracts; Mohamed Saleh provides Gold models to Basant; Mohsen validates outputs; Kerolos packages the stable stack; Amr coordinates acceptance.

Definition of Done

Code/docs are committed, acceptance criteria checked, errors/assumptions documented, and dependent members can run or consume the output. Dashboard calculations require reconciliation evidence; deployment requires repeatable instructions.

Risk Assessment & Mitigation Plan

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Probability (P) and Impact (I) are scored 1–5. Score=P×I. Levels: 1–4 Low, 5–9 Medium, 10–16 High, 17–25 Critical. Initial scores should be updated as evidence changes.

ID	Risk	P	I	Score	Mitigation / Contingency	Owner
R01	API outage	3	4	12 – High	Timeouts, retries, and logging; fall back to stable MVP endpoints.	Mohamed Ibrahim
R02	Schema/path changes	3	4	12 – High	Validate against the OpenAPI specification; pause and adapt the affected parser.	Mohamed Ibrahim
R03	Freshness target not feasible	4	3	12 – High	Measure source update cadence, revise the target if necessary, and display the last update time.	Amr Mahmoud
R04	Duplicate/inconsistent keys	4	4	16 – High	Use endpoint-specific keys and quarantine records with uncertain identities.	Mohamed Saleh
R05	Kafka loss/reprocessing	3	4	12 – High	Implement logging, retry and offset strategies, and idempotent writes where feasible.	Mohamed Ibrahim
R06	VM resource shortage	3	4	12 – High	Load-test and monitor resource usage; resize the VM or reduce the number of services if necessary.	Kerolos Adly
R07	Power BI connectivity/refresh limits	3	4	12 – High	Validate Power BI Desktop connectivity early; keep Power BI Service optional.	Basant Magdy
R08	Integration delays	3	4	12 – High	Agree on message contracts early and build a complete end-to-end vertical slice.	Amr Mahmoud
R09	Cloud costs exceed estimate	3	4	12 – High	Estimate and monitor cloud costs; stop unused resources.	Kerolos Adly
R10	Incorrect transformation metrics	3	5	15 – High	Implement dbt tests and reconciliation checks; disable any metric that produces incorrect results until fixed.	Mohamed Saleh
R11	Team delays	3	3	9 – Medium	Break work into small tasks, make blockers visible, and rebalance critical tasks when needed.	Amr Mahmoud
R12	Secrets committed	2	5	10 – High	Use environment variables and .gitignore; immediately revoke and rotate any exposed credentials.	Kerolos Adly
R13	Insufficient test time	4	4	16 – High	Freeze the MVP scope early and reserve sufficient time for integration and final testing.	Amr Mahmoud
R14	Unbounded raw storage	3	3	9 – Medium	Define a retention policy and archive or remove data according to documented rules.	Mohamed Saleh

Review risks weekly and after major assumption changes. Planned mitigation is not closure; confirm it has been tested.

Key Performance Indicators (KPIs)

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Important: Targets below are not achieved results. Mark each KPI Not Yet Measured until evidence exists.

KPI	Measurement	Initial Target	Evidence
End-to-End Freshness	Time from source data availability/update to a queryable Gold record.	5–10 minutes, where source update cadence permits.	Timestamped logs.
API Ingestion Success	Valid successful polls / attempted polls.	≥ 95%.	Producer logs.
Kafka Processing Success	Successfully persisted messages / consumed messages, accounting for retries.	≥ 99%.	Consumer logs and database checks.
Data Quality Pass Rate	Valid records / assessed records.	≥ 95% for agreed MVP rules.	Data validation report.
Gold Uniqueness	Number of unexplained duplicate business keys.	Zero unexplained violations.	dbt test output.
Mandatory dbt Tests	Passed mandatory tests / total mandatory tests.	100%.	dbt test output.
Reconciliation	Mandatory sample cases matching expected output.	100%.	Test checklist.
Dashboard Correctness	Critical dashboard measures reconciled against Gold data.	All critical measures pass validation.	Reconciliation evidence.
Dashboard Freshness	Time from Gold data update to updated data becoming visible in the report.	5–10 minutes, if supported.	Dashboard refresh test.
MVP Dashboard Coverage	Agreed dashboard requirements completed and validated.	100%.	Requirements traceability checklist.
Pipeline Availability	Successful health checks / total checks during the defined test window.	≥ 95% during the defined test window.	Health-check logs.
Docker Reproducibility	Successful runs in a clean, controlled environment.	3 consecutive successful runs.	Execution checklist.
Recovery Tests	Critical failure, restart, and retry scenarios passed.	100%.	Recovery test evidence.
Cloud Budget	Actual cloud cost compared with the approved budget limit.	Within the agreed limit.	Azure cost records.
Milestones	Milestones completed by their agreed dates.	On time or formally re-baselined.	Project board and milestone records.

For every KPI record date, conditions, actual result, pass/fail, evidence link, and corrective action. Never report a target as achieved before measurement.